

Haofei Liang

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Education

Sept. 2022 – Dec. 2026 (expected) Shanghai Jiao Tong University

Doctor of Engineering candidate in Computer Science and Technology.

Research area: applications and optimization of fully homomorphic encryption.

Advisor: Prof. Yu Yu

Sept. 2020 – Aug. 2022 Shanghai Jiao Tong University

Master's study in Computer Science and Technology; transferred to the Doctor of Engineering program in 2022.

Advisor: Prof. Yu Yu

Sept. 2015 – June 2019 Wuhan University

B.Sc. in Information Security.

Research Interests

My research focuses on **fully homomorphic encryption (FHE)**, especially the **TFHE** family of schemes. Specific topics include:

Functional and Programmable Bootstrapping Evaluating functions during bootstrapping while reducing latency.

Circuit Bootstrapping Efficient conversion between FHE ciphertext formats for deeper homomorphic computation.

Applications of FHE Privacy-preserving protocols including oblivious message retrieval and single secret leader election.

Publications

Conference Proceedings

- **Haofei Liang**, Zeyu Liu, Eran Tromer, Xiang Xie, and Yu Yu. *InstantOMR: Oblivious Message Retrieval with Low Latency and Optimal Parallelizability*. In: 35th USENIX Security Symposium (USENIX Security 2026), Baltimore, MD, USA, August 2026. (Published) [USENIX](#) · [ePrint: 2025/2317](#).
- **Haofei Liang**, Zeyu Liu, Yunhao Wang, Xiang Xie, Yu Yu, and Fan Zhang. *Relect: Single Secret Leader Election via FHE with Reduced Computation and Communication and Transparent Setup*. In: 33rd ACM Conference on Computer and Communications Security (ACM CCS 2026), The Hague, The Netherlands, November 15–19, 2026. (Accepted, to appear) [ACM CCS](#) · [ePrint: 2026/1619](#).
- Zhenkai Hu, **Haofei Liang**, Xiao Wang, Xiang Xie, Kang Yang, Yu Yu, and Wenhao Zhang. *Ajax: Fast Threshold Fully Homomorphic Encryption without Noise Flooding*. In: 35th USENIX Security Symposium (USENIX Security 2026), Baltimore, MD, USA, August 2026. (Published) [USENIX](#) · [ePrint: 2025/1834](#).
- Fengrun Liu, **Haofei Liang**, Xiang Xie, Yu Yu, Wenting Zheng, and Yuncong Hu. *HasteBoots: Proving TFHE Programmable Bootstrapping in Seconds*. In: 35th USENIX Security Symposium (USENIX Security 2026), Baltimore, MD, USA, August 2026. (Published) [USENIX](#) · [ePrint: 2025/261](#).

Conference Presentations

Aug. 2026 **USENIX Security 2026**, Baltimore, MD, USA – presented *InstantOMR*.

Nov. 2026 **ACM CCS 2026**, The Hague, The Netherlands – scheduled presentation of *Relect*.

Open-Source Projects

- **primus-fhe** – Open-source Rust library for high-performance TFHE research and implementation; lead developer and maintainer, responsible for the unified FHE core, Fourier/NTT arithmetic backends, core TFHE operations, and performance, correctness, and noise validation. Development is maintained in my repository and progressively integrated upstream into Primus Labs. **Current development branch** · **Upstream repository**.
- **InstantOMR** – Single-server oblivious message retrieval for recipient privacy in anonymous messaging. Its two-layer bootstrapping design combines TFHE and RLWE computation to reduce online latency and enable message-level parallelism; first author and lead implementation developer.
- **Relect** – Ring-LWE-based single-secret leader election using the algebraic structure of threshold FHE and tailored homomorphic circuits; removes trusted setup and supports dynamic election in each round; first author and lead implementation developer.
- **Lattice Estimator Web/API** – Wrote Python scripts to invoke lattice-estimator for concrete security evaluation of FHE parameters; building on those scripts, this project packages the evaluation workflow as a Docker-deployable Web/API tool for repeatable runs and result inspection.
- **Ajax** – Threshold key-generation and decryption protocols for FHEW-like schemes, using mask-then-open to avoid noise flooding; primarily responsible for implementing and optimizing the FHE component.
- **HasteBoots** – Succinct arguments tailored to TFHE programmable bootstrapping; primarily responsible for the FHE component, while collaborators developed the zero-knowledge proof component.

Skills

Cryptographic Research Long-term research in FHE/TFHE; familiar with LWE/RLWE, functional, programmable, and circuit bootstrapping, and threshold FHE; experience designing FHE-based privacy-preserving protocols and analyzing computation and communication costs; noise analysis, parameter selection, and concrete lattice-security evaluation; working knowledge of BFV, BGV, and CKKS.

Cryptographic Implementation and Optimization Rust development of FHE core libraries and research prototypes; implementation of TFHE/FHEW-like core operations, ciphertext transformations, and polynomial-arithmetic backends; developing a Fourier-domain path using open-source FFT backends; performance profiling, parallel optimization, benchmarking, and correctness/noise validation.

Research and Development Tools Python for experimental scripts, lattice-estimator invocation, data processing, and figure generation; able to read C reference implementations; Linux, Git, reproducible experiments, and academic writing workflows.

Languages Native Mandarin Chinese; English (CET-6; proficient reading and writing, conversational speaking).

Selected Links

- **Website:** haofeiliang.github.io
- **Publications:** haofeiliang.github.io/publications/
- **Projects:** haofeiliang.github.io/projects/
- **GitHub:** github.com/haofeiliang